Clàudia Janó Muñoz

PhD student - University of Cambridge

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Education

2022 - *est. Summer 2026*	**PhD in Physics**, *Search for exoplanets around M-dwarfs with the CMOS infrarred detector SPIRIT.*, Cavendish Laboratory, University of Cambridge. Supervised by Prof. Didier Queloz and Dr. Matthew Hooton.
2018 - 2022	**MPhys Physics with Astrophysics** First Class, University of Manchester.

Research Experience

Oct. 2021 - May 2022	**Masters project**, University of Manchester. Machine learning in spectra analysis from ALMA observations. Supervised by Prof. Gary Fuller.
Sep. 2021 - Oct. 2021	**Summer research project**, Universitat Autònoma de Barcelona. Simulating the effects of close stellar encounters with N-body simulators. Supervised by Dr. Ignasi Ribas.
Jun. 2021 - Aug. 2021	**UROP Summer Internship**, Imperial College London. Exoplanet atmospheric evolution with machine learning. Supervised by Dr. James Owen.

Publications

First and Co-Author

in prep.	**Janó Muñoz, C.**, et al. *Performance and prospects of the SPIRIT infrared CMOS detector for exoplanet detection*
June 2025	**Janó Muñoz, C.**, et al. *TOI-2407 b: a warm Neptune in the desert*, Monthly Notices of the Royal Astronomical Society, 541, 630-641
March 2023	Rogers, J. G., **Janó Muñoz, C.**, Owen, J. E., Makinen, T. L. *Exoplanet atmosphere evolution: emulation with neural networks*, Monthly Notices of the Royal Astronomical Society, 519, 6028-6043

Contributing Author

November 2025	Scott, M. G., et al. *Two temperate Earth- and Neptune-sized planets orbiting nearby fully convective M dwarfs*, Monthly Notices of the Royal Astronomical Society, submitted
October 2025	Barkaoui, K., et al. *TOI-7166 b: A Habitable Zone mini-Neptune planet around a nearby low-mass star*, Monthly Notices of the Royal Astronomical Society, accepted
October 2025	Zúñiga-Fernández, S., et al. *Two warm Earth-sized exoplanets and an Earth-sized candidate in the M5V-M6V binary system TOI-2267*, Astronomy & Astrophysics, 702, A85
August 2024	Zúñiga-Fernández, S., et al. *SPECULOOS: five years hunting terrestrial planets around ultra-cool dwarfs*, Proceedings of the SPIE, 13094
July 2024	Pedersen, P.P., et al. *Infrared photometry with InGaAs detectors: first light with SPECULOOS*, Proceedings of the SPIE, 13096
July 2024	Gillon, M., et al. *Detection of an Earth-sized exoplanet orbiting the nearby ultracool dwarf star SPECULOOS-3*, Nature Astronomy, 8, 865-878
October 2023	Triaud, A. H. M. J., et al. *An M dwarf accompanied by a close-in giant orbiter with SPECULOOS*, Monthly Notices of the Royal Astronomical Society: Letters, 525, L98-L104

Talks and posters

January 2026	*First Results from SPIRIT: A CMOS Detector for Exoplanets around Ultracool Dwarfs*, Rocky Worlds IV, Gröningen, Netherlands.
December 2025	*Discussion on SPIRIT: Infrared CMOS Photometry for Exoplanets around Ultracool Dwarfs*, LCLU discussion series, Cambridge, UK.
November 2025	*SPIRIT: A CMOS Detector for Exoplanets Around Ultracool Dwarfs*, Exoplanet Seminar, Observatoire de Paris-PSL, France.
November 2025	*Discovery of the Neptune-like planet TOI-2407 b*, Harding Scholars Internal Speaker Event, St Catherine's College, Cambridge, UK.
February 2025	*Exoplanet Detection with SPIRIT: Infrared CMOS Photometry and the Discovery of the Hot Neptune TOI-2407 b*, Hills Coffee Talks series, Battcock Centre for Experimental Astrophysics, Cambridge, UK.
February 2025	*TOI-2407b: A hot Neptune in the Neptune Desert discovered with the SPIRIT Infrared Detector*, Exoplanet Seminar series, University of Cambridge, UK.
June 2024	POSTER: *Ground-based transits in the infrared with SPIRIT*, Exoplanets V, Leiden, Netherlands.
January 2024	POSTER: *Finding transiting planets in the infrared with SPIRIT.*, Rocky Worlds III, ETH Zurich, Switzerland.
August 2022	*Emulating photoevaporation with Machine Learning*, University of California Santa Cruz, USA (remote).

Teaching Experience

Apr. 2024 - May 2024	**Part III Example Classes: Exoplanets Course** 2h per week, Cavendish Laboratory.
Nov. 2022 - May 2023	**Part IA Physics Supervision** 3h per week, Emmanuel College, University of Cambridge.

Observing Experience

Oct. 2022 - Jan. 2025	Bi-monthly remote observations with the Callisto and Ganymede telescopes from the SPECULOOS Southern Observatory.
Dec. 2023	On-site observations, maintenance and testing at the SPECULOOS Southern Observatory, Cerro Paranal, Chile.

Awards

October 2022	**Harding Distinguished Postgraduate Scholarship** (£20,000 per year of study), including a £2000 yearly stipend for academic travel.
December 2018	**Physics Success Scholarship** (£2000), for academic excellence in physics and maths.

Software

SMART	Co-author: Machine Learning tool for fitting proto-stellar spectra.

Extracurricular Activities and Outreach

Mar. 2025	**Open Day**, Volunteer, Institute of Astronomy, Cambridge.
Mar. 2025	**International Women's Day**, Running an exoplanets workshop for high school students. Institute of Astronomy, Cambridge.
Jan. 2025 - Apr. 2025	**International Women's Day Committee**, Member, Institute of Astronomy, Cambridge.
May 2021 - Feb. 2022	**Astronomy Society**, Chair, University of Manchester.
Feb. 2020 - May 2021	**Astronomy Society**, Science Officer, University of Manchester.

Skills

Programming	**Proficient**: Python, **Intermediate**: C++
Computing	**Experienced**: LaTeX, Linux, Docker, **Basic**: HPCs
Languages	**Proficient**: English **Native:** Catalan, Spanish **Intermediate**: German (B1)